

Rahul Reddy Puchakayala

AI Product Manager

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SUMMARY

AI Product Manager with 6+ years of experience defining AI platform strategy, building agentic systems, and operationalizing enterprise AI at scale. Led product initiatives supporting 800+ developers and 80+ AI use cases at Capital One, including AI governance, evaluation frameworks, RAG infrastructure, and model customization platforms. Proven track record of turning emerging AI capabilities into trusted production systems that improve decision quality, accelerate workflows, reduce operational risk, and deliver measurable business impact. Combines product strategy, technical depth, and cross-functional leadership across engineering, risk, legal, compliance, and executive stakeholders.

PROFESSIONAL EXPERIENCE

Capital One | *Product Manager, AI/ML Products*

January 2025 – Present

- Scaled enterprise AI platform to 800+ developers across 80+ use cases with 89% deployment coverage, by defining the 3-year product vision and aligning engineering, risk, legal, and compliance stakeholders around a unified AI Agent and Data platform roadmap
- Eliminated weeks of manual provisioning friction for engineering teams, by launching a zero-touch self-service VectorDB platform and AI prototyping environment enabling instant RAG application deployment
- Reduced analyst report generation time by 50% and query response times by 40%, by productizing LLaMA and GPT-4 RAG pipelines using hybrid ChromaDB and BM25 retrieval with cross-encoder reranking
- Scaled automated compliance monitoring from 1% to 100% of transactions while improving fraud detection accuracy by 25%, by deploying human-in-the-loop validation and NLP entity recognition as the core enterprise AI governance layer
- Established production AI governance standards across multiple business units, measured by quality, safety, groundedness, latency, and human-override metrics, by defining evaluation frameworks governing enterprise AI deployment decisions
- Eliminated \$1M+ in annual third-party software spend, by leading cross-functional programs across engineering, risk, legal, and compliance to replace vendor tools with internally built agentic GenAI solutions
- Standardized model customization, safety guardrails, and compliance evaluation across enterprise AI teams, by leading the LLM fine-tuning service from PRD through General Availability

Optum (UnitedHealth Group) | *Product Manager*

August 2021 – May 2024

- Reduced testing cycle times by 30% for 5,000+ enterprise healthcare engineers, by owning the product strategy for Scale Test Copilot and implementing AI-driven test prioritization and workflow automation
- Drove platform adoption from near-zero to 60% within six weeks, by redesigning anomaly detection with severity ranking and confidence scoring to shift from high-recall noise to actionable signal, cutting time to first successful API call from 5 days to 1 day and improving completion rates from 40% to 80%
- Improved cross-functional decision-making speed by 25%, by building product analytics frameworks tracking 10+ KPIs across platform health, adoption, and model performance to enable data-driven roadmap prioritization
- Influenced continued platform investment across a system supporting 5,000+ healthcare engineers, by presenting roadmap strategy, customer insights, and measurable business outcomes to senior leadership quarterly

Accenture | *Data Engineer*

January 2019 – July 2021

- Reduced pipeline integration times by 30% across multiple client engagements, by designing cross-source data integration architectures using Kafka, Spark, and Hive that resolved complex multi-team dependencies
- Ensured SLA compliance and enterprise reporting reliability across hundreds of millions of records, by building and scaling automated ETL pipelines processing structured and unstructured data at production scale
- Drove measurable revenue improvements across client portfolios, by analyzing large-scale streaming datasets on AWS to surface platform insights that directly informed business strategy and investment decisions

AI PRODUCT PROJECTS

RegulatedAI — Financial & Legal Intelligence Platform | 2026

- Reduced financial advisory turnaround from hours to under 15 seconds by designing an AI compliance workflow that enforces FINRA Rule 2111 and Reg BI requirements before any recommendation reaches the user, eliminating post-hoc audit liability
- Improved citation accuracy and reduced human overrides in legal AI outputs by designing a legal research product that verifies every citation against authoritative sources before delivery
- Reduced manual compliance review time by 85% by shipping a human approval workflow that automatically routes high-risk AI outputs to analyst review with full audit trail and decision logging

ProductOS — AI Operating System for Product Teams | 2026

- Accelerated product discovery and planning from weeks to under 30 seconds by designing an AI workflow that automated research synthesis, opportunity sizing, prioritization, and PRD generation from raw customer interviews and support tickets
- Improved the credibility and actionability of customer research outputs by introducing source attribution that renders findings as "8 of 12 interviews" rather than generic themes, giving PMs evidence they can act on without requesting further research
- Increased output quality and stakeholder trust by defining the full product brief structure before building any AI workflow, ensuring every output composes directly into a prioritized roadmap, PRD, or experiment plan

ExecutiveAI — AI Chief of Staff | 2026

- Reduced leadership follow-up overhead by 4-6 hours per week by designing an AI decision registry that automatically converts meeting transcripts into structured action items with owner assignment, priority levels, and escalation routing for critical decisions
- Eliminated ungoverned AI decision-making by designing an approval workflow where critical decisions automatically escalate to an executive queue with impact framing, AI recommendation, and immutable audit trail before any action is taken
- Gave leadership early visibility into execution risk by designing a week-over-week program health scoring product that synthesizes schedule risk, stakeholder sentiment, and decision velocity into a single trend metric before issues surface in status reports

Reknew AI — Search Infrastructure for Agentic Commerce | 2025

- Improved retrieval quality by 37% over keyword baselines across a 2,250-product catalog by leading the evaluation and selection of the optimal semantic search architecture, benchmarking 18 configurations using NDCG@10 and LLM-as-a-Judge frameworks

SKILLS

Product: Product strategy & roadmap ownership, PRDs & user stories, backlog prioritization, Agile (Scrum/Kanban), market & user research, KPI definition & analytics, go-to-market strategy, stakeholder alignment

AI & ML: LLMs (GPT-4, Claude, LLaMA), RAG systems (LangChain, LlamaIndex), TensorFlow, prompt engineering, NLP, XGBoost, vector search (ChromaDB), MLOps (Vertex AI, Kubeflow)

Builder Tools: Claude Code, Cursor, GitHub Copilot, Lovable

Technical: Python, SQL, REST APIs, ETL pipelines, CI/CD, GitHub Actions, Spark, Hadoop, Kafka, Hive

Tools and Platforms: AWS, GCP, Azure, Power BI, Tableau, Jira, Confluence, Figma, Aha!

EDUCATION

University of Texas at Dallas, U.S.A

May 2026

Master of Science, Information Technology & Management

- **Honors & Awards:** Dean's Excellence Scholarship recipient